

NEET-UG 2026 — Biology Categorization

Each of the 90 Biology questions (Q91–Q180) in the NEET-UG 2026 paper, mapped to its corresponding Unit / Subtopic in the official NEET-UG 2026 syllabus.

Sources: NEET-UG 2026 syllabus (NMC notification, Notice_20260108180635.pdf, Biology — 10 units) · Question paper + answer key + line-by-line explanations (questions on pp. 2–86, explanations on pp. 87–277).

Distribution across the syllabus units

Unit	Syllabus Unit	Q count	Question Numbers
1	Diversity in Living World	7	Q99, Q105, Q121, Q137, Q139, Q156, Q176
2	Structural Organisation in Animals and Plants	9	Q107, Q109, Q114, Q120, Q128, Q130, Q135, Q136, Q138
3	Cell Structure and Function	10	Q91, Q101, Q102, Q104, Q127, Q133, Q141, Q152, Q153, Q177
4	Plant Physiology	7	Q96, Q97, Q100, Q106, Q112, Q124, Q129
5	Human Physiology	12	Q140, Q143, Q145, Q146, Q148, Q150, Q159, Q161, Q168, Q172, Q173, Q180
6	Reproduction	8	Q108, Q115, Q117, Q142, Q151, Q154, Q163, Q169
7	Genetics and Evolution	14	Q92, Q94, Q95, Q103, Q111, Q119, Q131, Q157, Q158, Q160, Q166, Q174, Q175, Q179
8	Biology and Human Welfare	4	Q149, Q164, Q165, Q167
9	Biotechnology and Its Applications	8	Q98, Q110, Q113, Q123, Q132, Q144, Q147, Q171
10	Ecology and Environment	11	Q93, Q116, Q118, Q122, Q125, Q126, Q134, Q155, Q162, Q170, Q178
Total		90	

Class-level split (NCERT)

- **Class XI** units (1–5): ~45 questions — Diversity, Structure, Cell, Plant Physiology, Human Physiology
- **Class XII** units (6–10): ~45 questions — Reproduction, Genetics/Evolution, Human Welfare, Biotech, Ecology

Per-question categorization

Q#	Ans	Syllabus Unit	Subtopic	What the question tests (from the PDF explanation)
91	A	3 — Cell Structure and Function	Cell division — phases of the cell cycle	Match G1, S, G2, M phases to their key activities (growth, DNA replication, mitosis).
92	A	7 — Genetics and Evolution	Deviations from Mendelism — incomplete dominance, codominance, multiple alleles	Match each inheritance pattern with its classic example (e.g., incomplete dominance → Antirrhinum).
93	B	10 — Ecology and Environment	Biodiversity loss — habitat loss; extinction examples	Amazon deforestation for soy → habitat loss; Steller's sea cow and passenger pigeon hunted to extinction.
94	B	7 — Genetics and Evolution	Molecular basis of inheritance — transcription unit (promoter, structural gene, terminator)	Identify true statements about the three regions of a transcription unit and their orientation.
95	A	7 — Genetics and Evolution	Sex determination — haplodiploidy in honeybees	Fertilised diploid eggs → females; unfertilised haploid eggs → drones (males).
96	D	4 — Plant Physiology	Cellular respiration — site of glycolysis, ETS, TCA	Glycolysis → cytoplasm; ETS → inner mitochondrial membrane; TCA → mitochondrial matrix.
97	A	4 — Plant Physiology	Photosynthesis — Calvin cycle ATP/NADPH stoichiometry	Per glucose, Calvin cycle consumes 18 ATP and 12 NADPH (3 ATP + 2 NADPH per CO ₂ × 6).
98	A	9 — Biotechnology and Its Applications	Applications of biotechnology — GMOs, recombinant insulin, gene therapy	Match: GMO → Bt cotton; recombinant insulin → Humulin; etc.
99	B	1 — Diversity in Living World	Plant classification — gymnosperms (naked seeds)	Pinus is a gymnosperm; its ovules are borne exposed on megasporophylls (not enclosed in an ovary).
100	C	4 — Plant Physiology	Photosynthesis — Calvin cycle carboxylation enzyme	RuBisCO (Ribulose-1,5-bisphosphate carboxylase/oxygenase) catalyses CO ₂ fixation onto RuBP.
101	A	3 — Cell Structure and Function	Biomolecules — enzymes, alkaloids, lectins, drugs	Match trypsin (enzyme), morphine (alkaloid), concanavalin A (lectin), and a drug to their categories.
102	B	3 — Cell Structure and Function	Biomolecules — amino acid structure	Amino acids have an α -carbon bearing NH_2 , COOH , H , and an R-group — substituted methanes; zwitterions at neutral pH.
103	C	7 — Genetics and Evolution	Mendelian disorders — sickle-cell anaemia mutation	Point mutation in β -globin gene: Glu → Val at the 6th amino-acid position.
104	D	3 — Cell Structure and Function	Nucleus and nucleolus — site of rRNA synthesis	Nucleolus is the dense sub-nuclear body where rRNA is transcribed and ribosomal subunits are assembled.
105	D	1 — Diversity in Living World	Taxonomy — binomial nomenclature rules	Italics in print / underline when handwritten; genus capitalised; species lowercase; etc.
106	D	4 — Plant Physiology	Plant growth regulators — auxins, gibberellins, cytokinins, ethylene, ABA	Match 2,4-D (herbicide), GA3 (malting), cytokinin (cell division), ABA (dormancy) to functions.
107	A	2 — Structural Organisation in Animals and Plants	Plant anatomy — conjunctive tissue, Casparian strips, endodermis, pericycle	Match anatomical terms (conjunctive tissue between xylem & phloem; Casparian strips in endodermis) to their definitions.
108	D	6 — Reproduction	Sexual reproduction in flowering plants — pollination types (autogamy, geitonogamy, xenogamy)	Xenogamy alone introduces genetically different pollen (from a different plant) to the stigma.

Q#	Ans	Syllabus Unit	Subtopic	What the question tests (from the PDF explanation)
109	A	2 — Structural Organisation in Animals and Plants	Morphology — heterophylly and environmental (phenotypic) plasticity	Different leaf forms on the same plant in response to environmental cues = environmental / phenotypic plasticity.
110	A	9 — Biotechnology and Its Applications	Tools of recombinant DNA — restriction endonucleases	Identify which statements about restriction enzymes are NOT true ("molecular scissors", specific recognition sites, etc.).
111	B	7 — Genetics and Evolution	Molecular basis of inheritance — DNA fingerprinting steps	Correct order: isolate DNA + digest with restriction enzymes → electrophoresis → blotting → hybridisation with VNTR probe → autoradiography.
112	B	4 — Plant Physiology	Photosynthesis — PSII water splitting; C3 vs C4 pathway	Identify the incorrect photosynthesis statement (e.g., water splitting at PSII, C4 plants still use C3 cycle for biosynthesis).
113	B	9 — Biotechnology and Its Applications	Plant biotechnology — somatic hybridisation (protoplast fusion)	Order: isolate single cells → digest cell walls → isolate naked protoplasts → fuse → culture → regenerate hybrid plants.
114	C	2 — Structural Organisation in Animals and Plants	Plant anatomy — bulliform (motor) cells in grasses	Large thin-walled epidermal cells; turgid → leaf flat; flaccid → leaf rolls to reduce transpiration.
115	D	6 — Reproduction	Sexual reproduction in flowering plants — microsporogenesis sequence	Sporogenous tissue → pollen mother cells → microspore tetrads → pollen grains.
116	D	10 — Ecology and Environment	Biodiversity conservation — in situ vs ex situ	In situ = protection of species in their natural habitat (national parks, biosphere reserves); seed banks / safari parks are ex situ.
117	C	6 — Reproduction	Double fertilisation in angiosperms — ploidy of cells (n, 2n, 3n)	Egg/synergids/antipodals = n; zygote = 2n; primary endosperm nucleus / endosperm = 3n.
118	D	10 — Ecology and Environment	Sixth mass extinction — unprecedented current rate of species loss	Current extinction rate is several orders of magnitude higher than the geological background rate.
119	D	7 — Genetics and Evolution	Gene expression regulation — lac operon	lacZ → β-galactosidase; lacY → permease; lacA → transacetylase; lacI → repressor.
120	A	2 — Structural Organisation in Animals and Plants	Inflorescence — racemose (acropetal succession)	Main axis grows indefinitely; older flowers towards base, younger ones towards apex.
121	B	1 — Diversity in Living World	Whittaker's five-kingdom classification — criteria	Cell structure (pro/eu), body organisation (uni/multi), mode of nutrition, ecological role, phylogeny.
122	A	10 — Ecology and Environment	Biodiversity loss — 'Evil Quartet' (Wilson)	Habitat loss & fragmentation, over-exploitation, alien species invasion, co-extinctions.
123	D	9 — Biotechnology and Its Applications	Tools of biotechnology — PCR cycle (denaturation, annealing, extension)	Correct chronological steps of one PCR cycle: melt strands → anneal primers → Taq polymerase extends.
124	C	4 — Plant Physiology	Respiration — Respiratory Quotient (RQ)	RQ = volume of CO ₂ evolved / volume of O ₂ consumed; varies with substrate (carbs = 1, fats < 1, organic acids > 1).
125	D	10 — Ecology and Environment	Biodiversity — bioprospecting definition	Bioprospecting is the systematic exploration of molecular, genetic and species-level diversity for economically valuable products.
126	D	10 — Ecology and Environment	Ecosystem — decomposition terms (detritus, fragmentation, leaching, humification, mineralisation)	Match each decomposition term to its correct definition.
127	B	3 — Cell Structure and Function	Biomolecules — lipids, proteins (polypeptides), nucleic acids	Identify which biomolecule statements are correct: lipids hydrophobic; proteins = chains of amino acids; etc.

Q#	Ans	Syllabus Unit	Subtopic	What the question tests (from the PDF explanation)
128	C	2 — Structural Organisation in Animals and Plants	Plant anatomy — root zones (meristem, elongation, maturation / root hair)	Root hairs originate from the maturation (root-hair) zone of the root epidermis.
129	D	4 — Plant Physiology	Plant growth and development — phase of cell elongation	Elongation phase: rapid vacuolation, cell-wall loosening, increase in cell size — nuclei become less conspicuous, not more.
130	A	2 — Structural Organisation in Animals and Plants	Plant families — Solanaceae floral formula	Solanaceae: $\oplus \blacksquare K(5) C(5) A5 G(2)$ — actinomorphic, bisexual, pentamerous, syncarpous.
131	C	7 — Genetics and Evolution	Molecular basis — DNA packaging, histones, nucleosome	Histone octamer (2 each of H2A, H2B, H3, H4); histones are positively charged basic proteins; ~200 bp wraps each nucleosome.
132	C	9 — Biotechnology and Its Applications	Tools of biotechnology — cutting, separating, visualising DNA	Restriction enzymes cut DNA; agarose-gel electrophoresis separates fragments; ethidium-bromide / UV visualisation.
133	C	3 — Cell Structure and Function	Biomolecules — protein structure (primary, secondary α -helix, tertiary, quaternary)	Secondary structure = α -helix / β -sheet stabilised by H-bonds between backbone atoms.
134	A	10 — Ecology and Environment	Ecosystem — GPP, NPP, secondary productivity	GPP = total carbon fixed; NPP = GPP – R (energy available to consumers); secondary productivity = rate of consumer assimilation.
135	C	2 — Structural Organisation in Animals and Plants	Flower — types of placentation	Marginal → Pea; axile → China rose / lemon; parietal → mustard; free central → Dianthus; basal → sunflower.
136	C	2 — Structural Organisation in Animals and Plants	Frog anatomy — hepatic portal system, cranial nerves	Identify correct frog statements (hepatic portal vein takes blood from gut → liver; frog has 10 pairs of cranial nerves).
137	A	1 — Diversity in Living World	Class Aves — flightless bird with paddle-like forelimbs	Penguin: flightless bird whose wings are modified into flippers / paddles adapted for swimming.
138	D	2 — Structural Organisation in Animals and Plants	Frog — sexual dimorphism	Male frogs have vocal sacs and nuptial pads; "bulging eyes" is NOT a male-distinguishing feature.
139	A	1 — Diversity in Living World	Animal classification — Cyclostomata (Class Agnatha)	Cartilage endoskeleton, ecto-parasitic circular sucking mouth, 6–15 pairs of gill slits, no paired fins / scales → lamprey / hagfish.
140	C	5 — Human Physiology	Breathing and respiration — ordered steps	Pulmonary ventilation → gas exchange across alveoli → transport in blood → diffusion at tissues → cellular respiration.
141	D	3 — Cell Structure and Function	Cell organelles — non-membrane-bound, present in pro- & eukaryotes	Ribosomes are non-membrane-bound and present in both prokaryotes and eukaryotes.
142	C	6 — Reproduction	Reproductive health — assisted reproductive technologies (GIFT)	GIFT (Gamete Intrafallopian Transfer): transfer of gametes directly into the fallopian tube for in vivo fertilisation.
143	B	5 — Human Physiology	Locomotion and movement — mechanism of muscle contraction	Motor neuron → sarcolemma → sarcoplasmic reticulum releases Ca^{2+} → troponin–tropomyosin → cross-bridge cycle.
144	B	9 — Biotechnology and Its Applications	Cloning vectors — pBR322, BamHI, antibiotic-resistance selection	BamHI site in pBR322 lies within the tet gene → insertional inactivation → select on ampicillin, screen for tetracycline sensitivity.
145	A	5 — Human Physiology	Neural control and coordination — chemical synapse	Neurotransmitter binds receptors on the post-synaptic membrane (not the pre-synaptic membrane).

Q#	Ans	Syllabus Unit	Subtopic	What the question tests (from the PDF explanation)
146	B	5 — Human Physiology	Locomotion and movement — axial skeleton (skull, vertebrae, ribs)	Human skull is dicondylic; 7 cervical vertebrae; rib classification (true / false / bicephalic) — evaluate which statements are correct.
147	B	9 — Biotechnology and Its Applications	Applications in health — α -1 antitrypsin for emphysema	Recombinant α -1 antitrypsin is used to treat hereditary emphysema (genetic AAT deficiency).
148	C	5 — Human Physiology	Body fluids and circulation — Rh blood grouping, erythroblastosis foetalis	EBF occurs when an Rh– mother carries an Rh+ fetus (after first sensitisation); identify incorrect Rh-compatibility claims.
149	A	8 — Biology and Human Welfare	Drug and alcohol abuse — effects of nicotine, opioids, cannabinoids, cocaine	Match each drug to its primary physiological effect (e.g., nicotine → stimulant via catecholamines).
150	B	5 — Human Physiology	Locomotion and movement — muscular/skeletal disorders	Tetany — muscle spasms from low Ca^{2+} ; arthritis — joint inflammation; osteoporosis; myasthenia gravis; muscular dystrophy.
151	C	6 — Reproduction	Reproductive health — contraceptive methods (IUDs)	Progestasert → hormone-releasing IUD; Multiload 375 → Cu-releasing IUD; LNG-20; CuT etc.
152	D	3 — Cell Structure and Function	Cell membrane — fluid-mosaic model, lipid bilayer, proteins	Identify correct membrane statements (~52% protein in RBC membrane, phospholipid bilayer); mesosomes belong to prokaryotes.
153	D	3 — Cell Structure and Function	Cell organelles — endomembrane system (ER, Golgi, lysosomes, vesicles)	Endomembrane system: nuclear envelope, ER, Golgi, lysosomes, vesicles — mitochondria, plastids, peroxisomes are NOT part of it.
154	A	6 — Reproduction	Human reproduction — embryonic developmental milestones (gestational weeks)	First movement & hair on head → ~20 weeks; eyelashes by ~24 wks; full term → ~38–40 wks.
155	C	10 — Ecology and Environment	Population interactions — sexual deceit (orchid – wasp mimicry)	Mediterranean orchid <i>Ophrys</i> mimics female bee/wasp to attract males for pollination — pseudo-copulation → sexual deceit.
156	A	1 — Diversity in Living World	Animal classification — Class Osteichthyes (bony fishes)	Identify the option containing only bony fishes (e.g., Catla, Labeo, sea horse); exclude Chondrichthyes / non-fish.
157	A	7 — Genetics and Evolution	Biological evolution — geological timeline of life	65 mya → K-Pg extinction (dinosaurs); 500 mya → invertebrate radiation; 200 mya → mammals appear; etc.
158	B	7 — Genetics and Evolution	Sex determination / gametogenesis — haploid gametes by mitosis (honeybee drones)	Drones are haploid (parthenogenetic) and produce sperm by mitotic divisions, not meiosis.
159	D	5 — Human Physiology	Body fluids and circulation — differential WBC count	Given 8000 WBC/mm ³ with eosinophils ~1–6% and lymphocytes ~20–45%, compute approximate counts.
160	D	7 — Genetics and Evolution	ABO blood group — multiple alleles, Mendelian probability	Heterozygous A (IA i) × heterozygous B (IB i) → 1 AB : 1 A : 1 B : 1 O → probability of given offspring blood group.
161	A	5 — Human Physiology	Excretion — renin–angiotensin–aldosterone system (RAAS)	Order: ↓BP/GFR → renin from JGA → angiotensin I → II → vasoconstriction + aldosterone → Na ⁺ /H ₂ O reabsorption.
162	A	10 — Ecology and Environment	Population interactions — mutualism, parasitism, amensalism, predation, competition	Evaluate statements describing each interaction (mutualism +/-, parasitism +/-, amensalism 0/-, etc.).
163	A	6 — Reproduction	Human reproduction — spermatogenesis	Spermatogonia divide mitotically → primary spermatocyte (2n) → meiosis I → secondary spermatocyte (n) → meiosis II → spermatids.

Q#	Ans	Syllabus Unit	Subtopic	What the question tests (from the PDF explanation)
164	A	8 — Biology and Human Welfare	Human disease — Plasmodium life cycle (malaria)	Sequence in human host: sporozoites injected by mosquito → liver (exo-erythrocytic) → RBC → trophozoite → schizont → merozoite → gametocytes.
165	C	8 — Biology and Human Welfare	Microbes in human welfare — microbial bioactive molecules	Streptokinase → dissolve clots; statins → cholesterol-lowering; cyclosporin A → immunosuppressant; LAB → curd.
166	C	7 — Genetics and Evolution	Evolution — convergent vs divergent (homologous vs analogous structures)	Whale and bat forelimbs are homologous (divergent evolution) — NOT an example of convergent evolution.
167	D	8 — Biology and Human Welfare	Microbes in food processing — Swiss cheese	Propionibacterium sharmanii ferments lactic acid producing CO ₂ → large holes in Swiss cheese.
168	D	5 — Human Physiology	Chemical coordination — adrenal hormones (cortisol, aldosterone)	Cortisol → anti-inflammatory / stress response; aldosterone → Na ⁺ and water reabsorption from renal tubules.
169	B	6 — Reproduction	Human reproduction — layers around the ovum	Outer → inner: corona radiata → zona pellucida → plasma membrane of the egg.
170	B	10 — Ecology and Environment	Populations — Verhulst–Pearl logistic growth	$dN/dt = rN \times (K - N)/K$, where K is the carrying capacity (sigmoid curve).
171	A	9 — Biotechnology and Its Applications	Applications in agriculture — Bt crops, Cry proteins	cryI _{Ac} → cotton bollworms; cryII _{Ab} → corn borer; cryI _{Ab} similar; specific Cry proteins target specific insect orders.
172	D	5 — Human Physiology	Excretion — Juxta Glomerular Apparatus (JGA)	JGA components: macula densa (DCT) + juxtaglomerular cells (afferent arteriole) + extraglomerular mesangial cells.
173	B	5 — Human Physiology	Respiration — modes of breathing across animals	Molluscs → branchial (gills); Reptiles → pulmonary only; Adult amphibians → pulmonary + cutaneous; Amoeba → cellular / by diffusion.
174	B	7 — Genetics and Evolution	Genetic code — sickle-cell anaemia mutant codon	Normal codon GAG (Glu) → mutant GUG (Val) at the 6th position of β-globin → HbS → polymerisation and sickling.
175	C	7 — Genetics and Evolution	Sex determination — XO system (grasshopper)	Female grasshopper = XX (2n + 2 chromosomes); male = XO (2n + 1) — one fewer chromosome than the female.
176	B	1 — Diversity in Living World	Animal kingdom — characteristic features (Platyhelminthes, Echinodermata, etc.)	Platyhelminthes → incomplete digestive system (true); adult echinoderms → radial symmetry (so claim of bilateral is false).
177	C	3 — Cell Structure and Function	Biomolecules — enzyme classification (IUBMB classes)	Reaction cleaves C–C bond, forms C=C without hydrolysis / oxidation → enzyme is a LYASE.
178	B	10 — Ecology and Environment	Ecosystem — ecological pyramids (number, biomass, energy)	Pyramid of energy is always upright; biomass / number pyramids can be inverted (e.g., parasite trees, marine ecosystems).
179	A	7 — Genetics and Evolution	Human evolution — ancestral sequence	Ramapithecus → Australopithecus → Homo habilis → Homo erectus → Neanderthal → Homo sapiens.
180	B	5 — Human Physiology	Breathing — respiratory volumes (TV, IRV, ERV, RV)	Tidal Volume ≈ 500 mL; IRV ≈ 2500–3000 mL; ERV ≈ 1000–1100 mL; Residual Volume ≈ 1100–1200 mL.

Notes on borderline / cross-listed items

A few questions sit at the boundary of two syllabus units. These are the choices made and why:

Q109 (heterophylly)

Sits between Unit 2 (Structural Organisation — leaf morphology) and Unit 4 (Plant Physiology — growth/development). Classified under Unit 2 since the question is fundamentally about morphological variation; the developmental explanation is secondary.

Q111 (DNA fingerprinting)

NCERT places DNA fingerprinting under the Molecular Basis of Inheritance chapter (Unit 7), even though it uses recombinant-DNA tools that overlap with Unit 9.

Q117 (ploidy in angiosperms)

Question is about cell ploidy during double fertilisation — placed under Unit 6 (Reproduction); ploidy as a concept also belongs to Unit 3 (Cell Cycle).

Q125 (bioprospecting)

Bioprospecting is taught in NCERT under the Biodiversity & Conservation chapter (Unit 10). Has obvious overlap with Unit 9 (biotech applications).

Q147 (α -1 antitrypsin)

Classified under Unit 9 (Biotechnology in Health) since the question is about a recombinant therapeutic protein. The underlying disease (emphysema) belongs to Unit 5.

Q165 (streptokinase / statins)

Both are microbial-derived bioactive molecules; NCERT covers them under "Microbes in Human Welfare" → Unit 8.

Q173 (modes of respiration)

Crosses Unit 1 (animal diversity) and Unit 5 (Breathing & Respiration). Placed under Unit 5 because the question is fundamentally about respiratory mechanisms.

Q177 (lyase enzyme class)

Enzyme nomenclature/classification is in NCERT under Biomolecules (Unit 3 — Cell Structure and Function).